Quenching Timescales of Dwarf Satellites in Cosmological Simulations of Milky Way-Mass Galaxies

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Introduction

Observations have revealed a dichotomy between satellite and field dwarf galaxies with $M_* = 10^{6\text{-}9}\,M_\odot$: satellites are preferentially **quenched** [4].

Many processes have been proposed to explain this dichotomy:

» **strangulation** (reduced gas accretion) may quench high-mass dwarfs ($M_* > 10^8\,M_\odot$) [14]

» direct gas-removal by **ram pressure stripping** is generally thought to be the dominant mechanism quenching intermediate-mass dwarfs [3, 9], and may be sped up by **stellar feedback** processes [1]

These processes are also thought to operate at very different timescales:

» **strangulation** typically takes 8-10 Gyr to quench a satellite following infall

» **ram pressure stripping** typically takes 1-2 Gyr to quench a satellite following infall

Thus, the variation in timescale can indicate the dominant process at play. In this project, we use ultra-high resolution simulations to explore the mass-dependence of dwarf satellite quenching. Specifically, we determine the times of quenching and infall to assess the efficiency of environmental quenching at different stellar masses, and compare to observations.

Simulations

We use the recently completed DC Justice League suite of **cosmological zoom-in simulations** generated from the ChaNGa tree+SPH code [11].

ChaNGa includes physics such as the cosmic UV background [6], "blastwave" SN feedback, and low-temperature metal cooling.

The 4 Justice League simulations are centered on Milky Way-mass halos and their surrounding satellite population, and are run at **ultra-high resolution** (3300/994 $M_\odot$ gas/star particles), allowing us to resolve halos down to $10^6\,M_{vir}$.

Sim	M_{vir} ($M_\odot$)	M_* ($M_\odot$)	M_{gas} ($M_\odot$)	R_{vir} (kpc)
Sandra	2.7×10^{12}	1.9×10^{11}	2.0×10^{11}	432
Ruth	1.3×10^{12}	1.0×10^{11}	9.3×10^{10}	340
Sonia	1.2×10^{12}	9.0×10^{10}	9.7×10^{10}	333
Elena	8.3×10^{11}	9.0×10^{10}	3.5×10^{10}	293

Table 1: Host halo properties.

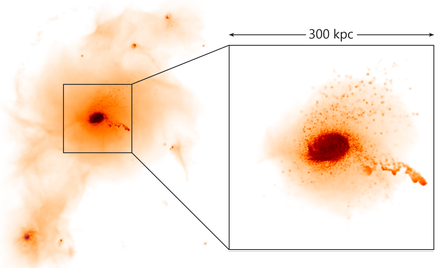


Figure 1 (left): Gas density distribution of Ruth.

At What Mass Scales Do Satellites Quench?

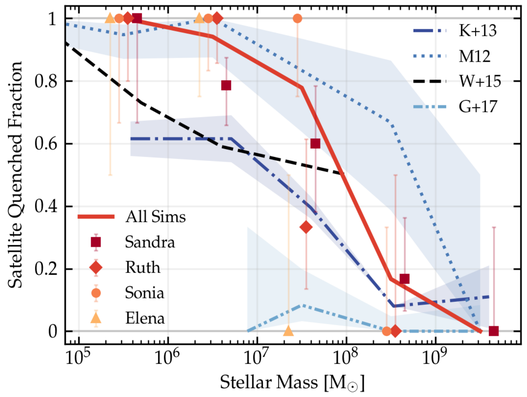
Figure 2 shows the quenched fraction vs. stellar mass. We see considerable variation across simulations, particularly at $M_* \sim 10^{7\text{-}9}\,M_\odot$, but our simulations match most observations in the aggregate. However, the **SAGA survey** [5] finds the majority of satellites to be star-forming, even at low masses ($10^6\,M_\odot$). The inconsistency between SAGA and our results may indicate that environmental quenching is generally less common than we model and observe in the Local Group.

Results

» Quenched fractions for simulated galaxies approximately replicate those observed

» While most satellites with $M_* < 10^7\,M_\odot$ are quenched, most with $M_* > 10^8\,M_\odot$ are star-forming

» Notable discrepancy with the SAGA survey



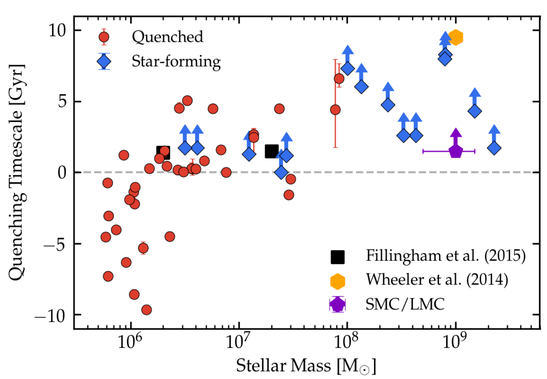
▲ **Figure 2** (above)**:** Fraction of satellites that are quenched at $z = 0$, in 1 dex bins of stellar mass. Data for each simulation are shown separately, with the aggregate for all simulations shown as a solid red line. Observational data are from [10], [8], [13], and [5]. Error bars show 68% uncertainty in the binomial proportion.

When Do Satellites Quench?

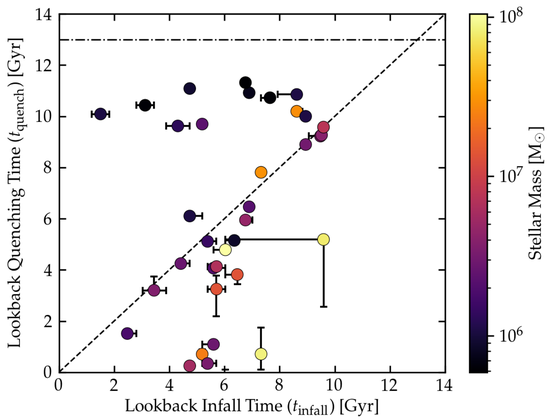
Figure 3 (right) shows the quenching time vs. the infall time of satellite galaxies quenched at $z = 0$. It shows two distinct groups of satellites

1. those that quench early (t_{quench} > 9 Gyr) with little correlation to infall time
2. those that quench right at infall (along the diagonal line)

Figure 4 (below) shows the quenching timescale, or the time after infall until a satellite quenches, vs. stellar mass. It shows considerable variation in the timescale across mass, with many intermediate mass ($10^{6\text{-}8}\,M_\odot$) satellites having rapid timescales. Massive satellites ($>10^8\,M_\odot$) have longer quenching timescales, and are unviersally star-forming.



▲ **Figure 4** (above)**:** Satellite quenching timescale (t_{infall} - t_{quench}) vs stellar mass. Positive values indicate quenching after infall. Satellites that remain star-forming at $z = 0$ are represented here as lower limits on their potential quenching timescale. Observed timescales from [3] and [14] are shown, as well as a lower limit for the SMC/LMC system from [7].



▲ **Figure 3** (above)**:** Lookback quenching time (t_{quench}) vs lookback infall time (t_{infall}) for quenched satellites surviving to $z = 0$. The quenching time is defined as the lookback time at which a galaxy last crossed below sSFR = 10^{-11} yr^{-1}. The infall time is defined as the lookback time to the halo's first crossing of its host's virial radius. For both, error bars indicate sensitivity to threshold choice. Points to the right of the dotted line quench after infall, and the dot-dash line indicates the approximate time of the end of reionization.

Results

» Satellites with $M_* \sim 10^{6\text{-}8}\,M_\odot$ typically have rapid quenching timesales of 1-2 Gyr, but up to 5 Gyr

» Massive satellites with $M_* > 10^8\,M_\odot$ have much larger potential timescales, up to 8 Gyr

» We see a stellar mass threshold of $10^8\,M_\odot$ above which infalling dwarf galaxies are more resistant to environmental quenching and quenching timescales become much longer

Conclusions

The goal of this project was to measure the quenched fractions and quenching timescales of satellites in the JL simulations, to provide insight into the relative importance of quenching mechanisms. We conclude the following:

1. Quenched fractions vary widely over stellar mass, following observations of the Local Group
2. Many intermediate-mass satellites have rapid quenching timescales of 1-2 Gyr, likely from ram pressure aided by stellar feedback outflows
3. Above a critical mass threshold of $M_* \sim 10^8\,M_\odot$ satellites are largely resistant to environmental quenching, or have long quenching timescales. This threshold is consistent with literature [3, 12, 14] and suggests that strangulation might be the dominant quenching mechanism at this mass.

The results of the SAGA survey [5] are particularly interesting in the context of quenching timescales. SAGA satellites might be more similar to those above our mass threshold: recently infalling and/or massive enough to resist rapid quenching by ram pressure stripping.

Future Work

» Using gas-depletion timescales to better constrain the role of strangulation both at high and intermediate masses

» Analysis of observational signatures of different quenching mechanisms, such as changes in metallicity and gas inflow/outflow rates

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